

LAB ASSIGNMENT

C Programming — 10 Independent Assignments (Data Types, Operators, if-else, else-if, switch)

Assignment 1: Data Types and Formatted I/O (10 Marks)

Topic / Reference: Data types (int, float, double, char) and formatted I/O using scanf()/printf(). Refer to: C Data Types & Variables; printf/scanf format specifiers (%d, %f, %lf, %c).

1. Write a C program that declares variables of type int, float, double, and char. Read a student's roll number, height (in meters), CGPA, and grade (a single character) from the user, and display them back using appropriate format specifiers with two decimal places for the float/double values.

Assignment 2: Arithmetic and Assignment Operators (10 Marks)

Topic / Reference: Arithmetic operators (+, -, *, /, %) and compound assignment operators (+=, -=, *=, /=). Refer to: Operators in C — Arithmetic and Assignment Operators.

2. A grocery store sells rice at a given price per kg. Write a program that accepts the price per kg and the quantity purchased (in kg), then computes the total cost, applies a flat 5% discount, and adds a 2% packaging charge on the discounted amount. Print the final payable amount. Use compound assignment operators (+=, -=, *=) at least twice.

Assignment 3: Relational and Logical Operators (10 Marks)

Topic / Reference: Relational operators (>, <, ==, !=, >=, <=) and logical operators (&&, ||, !). Refer to: Operators in C — Relational and Logical Operators.

3. Write a program that accepts three integers and, without using if statements, prints the result (1 for true, 0 for false) of the following expressions: (a) whether all three numbers are equal, (b) whether at least one number is negative, (c) whether the first number is greater than both the other two. Use only relational and logical operators.

Assignment 4: Simple if-else (10 Marks)

Topic / Reference: Simple if-else statement (single condition, two-way branching). Refer to: Control Statements — if, if-else.

4. Write a program to accept the length of three sides of a triangle and check whether a valid triangle can be formed (sum of any two sides must be greater than the third). If valid, further check and print whether the triangle is equilateral, isosceles, or scalene.

Assignment 5: else-if Ladder (10 Marks)

Topic / Reference: else-if ladder (multi-way branching with ordered conditions). Refer to: Control Statements — else-if Ladder.

5. An electricity board charges domestic consumers as follows: first 100 units at Rs. 1.50/unit, next 100 units at Rs. 2.50/unit, next 100 units at Rs. 4.00/unit, and beyond 300 units at Rs. 6.00/unit. Write a program that reads the number of units consumed and computes the total electricity bill using an else-if ladder.

Assignment 6: switch Statement (10 Marks)

Topic / Reference: switch-case statement and the break keyword. Refer to: Control Statements — switch-case.

6. Write a menu-driven program using switch that accepts two numbers and an operator choice (1 = Addition, 2 = Subtraction, 3 = Multiplication, 4 = Division) and displays the result. Handle division by zero and invalid menu choices appropriately.

Assignment 7: Nested if-else with switch (10 Marks)

Topic / Reference: switch-case combined with nested if-else inside each case. Refer to: Control Statements — switch-case and Nested if-else.

7. A college classifies student results using a category code entered by the user: 1 = Science, 2 = Commerce, 3 = Arts. Based on the category (using switch) and the marks obtained (using nested if-else within each case), print the grade as per the following rule: Science/Commerce — A (≥ 75), B (60-74), C (40-59), Fail (< 40); Arts — A (≥ 70), B (55-69), C (35-54), Fail (< 35).

Assignment 8: Combining Operators with else-if (10 Marks)

Topic / Reference: Arithmetic operators combined with an else-if ladder for tiered/conditional calculations.

Refer to: Operators + Control Statements — else-if Ladder.

8. A cab booking service charges a base fare of Rs. 50, plus Rs. 12 per km for the first 10 km, Rs. 10 per km for the next 15 km, and Rs. 8 per km beyond that. Additionally, if the booking is made between 11 PM and 5 AM (accept the hour as a 24-hour integer input), a night surcharge of 25% is applied to the total fare. Write a program to compute and display the final fare.

Assignment 9: Tiered Logic Bridge Problem (10 Marks)

Topic / Reference: else-if ladder combined with switch-case in the same program (tiered rates + category selection). Refer to: Control Statements — else-if Ladder and switch-case (combined use).

9. A laptop manufacturing company pays its salespersons a base salary of Rs. 1500, plus a bonus per laptop sold and a commission on total sales. Write a program where: the bonus per laptop depends on the quantity sold — Rs. 150 if quantity < 20 , Rs. 200 if quantity is between 20 and 40 (inclusive), and Rs. 250 if quantity > 40 . Separately, the commission rate depends on a product category code entered by the user (1 = Laptop $\rightarrow$ 2%, 2 = Tablet $\rightarrow$ 1.5%, 3 = Accessory $\rightarrow$ 1%), selected using switch. Compute and display the gross salary.

Assignment 10: Final Problem (10 Marks)

Topic / Reference: Cumulative application of data types, arithmetic operators, and conditional statements.

Refer to: All previous exercises — Data Types, Operators, if-else/else-if/switch (combined application).

Bloom's Taxonomy Level: Create (design and construct a complete, working salary-computation program from a real-world specification)

A laptop manufacturing company has the following monthly compensation policy for its salespersons:

- Minimum base salary: Rs. 1500.00
- Bonus for every laptop sold: Rs. 200.00
- Commission on the total monthly sales: 2%

Since laptop prices change, the sale price per laptop is fixed at the start of every month. Write a C program that accepts the number of laptops sold and the price per laptop, then computes and displays the gross salary using:

$$\text{Gross Salary} = \text{Base Salary} + (\text{Quantity} \times \text{Bonus Rate}) + (\text{Quantity} \times \text{Price}) \times \text{Commission Rate}$$